

TUSHAR PRAKASH

Delhi, India

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EDUCATION

Delhi Technological University (Formerly DCE), Delhi, India

Aug'19 - May'23

Bachelor of Technology (B.Tech), Department of Electronics and Communication Engineering

GPA: 8.58/10.0 (3.72/4.0)

Coursework: Computer Vision, Neural Networks, Advanced Data Structures, Algorithm Analysis and Design, Software Engineering and Digital Signal Processing.

PUBLICATIONS

Conferences

- **T. Prakash**, B. Singh and N. Pedanekar, N. Chaturvedi. *Obliviate: Efficient Unlearning in Recommender Systems*. In Proceedings of the International Conference on Machine Learning (**ICML**) **2026**. 
- O. Susladkar, **T.Prakash**, I. Lourentzou, *et.al.* *Best of Both Worlds: Multimodal Reasoning and Generation via Unified Discrete Flow Matching*. In Proceedings of the International Conference on Machine Learning (**ICML**) **2026**. 
- O. Susladkar, **T.Prakash**, I. Lourentzou, *et.al.* *PyraTok: Language-Aligned Pyramidal Tokenizer for Video Understanding and Generation*. In Proceedings of the IEEE Computer Vision and Pattern Recognition (**CVPR**) **2026**. 
- O. Susladkar, D. H. Jang, **T.Prakash**, I. Lourentzou, *et.al.* *RewardFlow: Generate Images by Optimizing What You Reward*. In Proceedings of the IEEE Computer Vision and Pattern Recognition (**CVPR**) **2026**. 
- **T.Prakash**, O. Susladkar, I. S. Dhillon and S. Mittal. *Pyramidal Spectrum: Frequency-based Hierarchically Vector Quantized VAE for Videos*. In Proceedings of the Winter Conference on Applications of Computer Vision (**WACV**), **2026**. 
- **T.Prakash**, R. Jalan, B.Singh and N. Pedanekar. *Dynamic Task-adaptive Meta Optimization for User Cold-Start Recommendation*. In Proceedings of the 28th European Conference on Artificial Intelligence (**ECAI**) **2025**. 
- **T.Prakash**, R. Jalan, N. Onoe. *MVBN: Enhancing Social Recommendation with Multi-View BERT Network*. In Proceedings of the 23rd IEEE International Conference on Data Mining (**ICDM**), **Shanghai, China, 2023**. 
- **T.Prakash**, R. Jalan, B. Singh, N. Onoe. *CR-SoRec: BERT driven Consistency Regularization for Social Recommendation*. In Proceedings of the 17th ACM Conference on Recommender Systems (**RecSys**) **2023**.  
- **T.Prakash**, T. Dhamija, R. Kumar, J. Panda. *Leveraging Explainable Artificial Intelligence for Understanding the Effect of Model Capacity on Training Dataset Size*. In Proceedings of the 17th IEEE International Conference on Service Operations and Logistics, and Informatics (**SOLI**) **2022**. 

Journals

- I. Joshi and **T. Prakash**, R. Kumar and A. Dantcheva. *Unsupervised Domain Alignment of Fingerprint Denoising Models Using Pseudo Annotations*. Published in **Springer's Multimedia Tools and Applications (Q1)**, **2023**. 
- I. Joshi, **T. Prakash**, R. Kumar, A. Dantcheva, S. Dutta Roy, P. K. Kalra. *Context-Aware Restoration of Noisy Fingerprints*. Published in **IEEE Sensors Letters**, **2022**. 

Workshops

- R. Jalan, **T. Prakash**, N. Pedanekar. *LLM-BRec: Personalizing Session-based Social Recommendation with LLM-BERT Fusion Framework*. In Proceedings of the Generative Information Retrieval Workshop, 47th ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR-Workshop**), Washington D.C., USA, 2024. [↗](#)

Patents

- **T. Prakash**, B. Singh and N. Pedanekar. System and Method for Generative Entertainment Content Creation Using Large Language Models and Evolutionary Optimization. **Patent in filing**.
- R. Jalan, **T. Prakash**, and N. Onoe. *Social Network Information-Based Recommendations Using a Transformer Model*. United States Patent Application **US20240331008A1**, Filed: March 7, 2024, Published: October 3, 2024, Assigned to Sony Group Corporation. Status: Pending. [↗](#)

RESEARCH/WORK EXPERIENCE

Machine Learning Consultant, **Sony Research India**, Bengaluru, India Jul'23 - Present
Data Science Team

- Built an 8B-parameter unified generative recommender powering multiple surfaces at SonyLIV¹ (such as next-item prediction, tray ranking, spotlight, related-content). Trained from scratch on 40B behavioral tokens via a three-stage pipeline (continual pretraining → multi-task SFT → GRPO RL). Introduced hierarchical Semantic-ID tokenization representing items directly in the LLM embedding space, replacing large user/item embedding tables and cutting embedding memory by 95%. Outperformed SOTA recommenders by 17.4% on ranking and retrieval in A/B tests over 72 Million monthly active users.
- Developed a personalized promo-generation model to boost engagement with new titles on SonyLIV. Built atop the 8B recommender, trained a flow-matching diffusion model conditioned on each user's historical preferences, item metadata, and time-stamped plot synopses and key frames, generating per-user promo creative. Lifted CTR on new content by 67%.
- Accelerated training throughput $3\times$ (1.7M → 5.4M user-batches per GPU) via deferred parameter initialization, and scaled the pipeline to 70M+ users with distributed training.
- Proposed novel methods for social, session-based, cold-start, and unlearning recommendation (multi-task collaborative filtering, consistency regularization, dynamic graph modeling), validated on public benchmarks and SonyLIV data with up to 46.6% gains over SOTA baselines; published at RecSys'23, ICDM'23, ECAI'25, and ICML'26.

Research Intern, **University of Alberta**, Edmonton, Canada Jun'22 - Aug'22
Mitacs GRI'22

- Improved the robustness of vision models against adversarial attacks by removing black-box noise perturbations from the CIFAR dataset.
- Designed and implemented a reverse knowledge distillation framework with multi-layer feature alignment for adversarial denoising, improving PSNR by 5.5% over SOTA on CIFAR under black-box attacks.
- Conducted the project as part of the Mitacs Globalink Research Internship (GRI'22) program at the University of Alberta, Canada.

Software Engineer Intern, **Microsoft**, Bengaluru, India Apr'22 - May'22
Azure Global Infrastructure Team

- Designed and implemented an automated API testing framework for 5G network infrastructure, enabling large-scale validation of distributed services.
- Developed and deployed multiple PyATS-based testbeds to verify end-to-end communication among interdependent APIs within a Kubernetes microservice environment, improving testing coverage and reliability.

Research collaboration, **INRIA Sophia Antipolis**, France Nov'21 - Apr'22
STARS Research Team

¹SonyLIV is Sony Pictures' OTT platform with 100M+ monthly active users

- Implemented novel multi-task learning and self-supervised learning techniques to address domain shift in fingerprint denoising models.
- Proposed two approaches: (1) a multi-task framework with a self-supervised image rotation auxiliary task, improving quality scores by 8.60% over SOTA, published in *Multimedia Tools and Applications*; and (2) a self-supervised jigsaw puzzle task, yielding a further 5.83% enhancement in denoising performance, published in *IEEE Sensors Letters*.

EXTRA-CURRICULAR ACTIVITIES

Teaching Experience: Served as a Teaching Assistant for *EC 311: Algorithm Design and Analysis* (Aug'22 – Dec'22), assisting in coursework, evaluation, and tutorial sessions for 120+ undergraduate students.

Community Outreach: Contributed to the “*Desh Ke Mentor*” state government initiative, mentoring high school students from underrepresented groups and introducing them to STEM education and career pathways.

ACHIEVEMENTS

🏆 Best Paper Award: Received the Best Paper Award for *CR-SoRec: BERT-driven Consistency Regularization for Social Recommendation* at Sony Research India’s Annual Research Conference 2023.

Mitacs Globalink Research Internship (2022) 📄 : Selected for the prestigious Mitacs GRI'22 Fellowship, a fully funded in-person research internship at the University of Alberta, Canada.

TECHNICAL SKILLS

Programming: Python, JavaScript, C/C++.

Frameworks: PyTorch, TensorFlow, JAX.

Libraries & Tools: OpenCV, scikit-learn, NumPy, PyATS, MLflow, LangChain, Ray, Kubernetes.